WGS Quality Control Report

Batch Name: 2026-09-25_P

Experiment Name: 26ARS_TRICYCLE_HIN_MCM7W

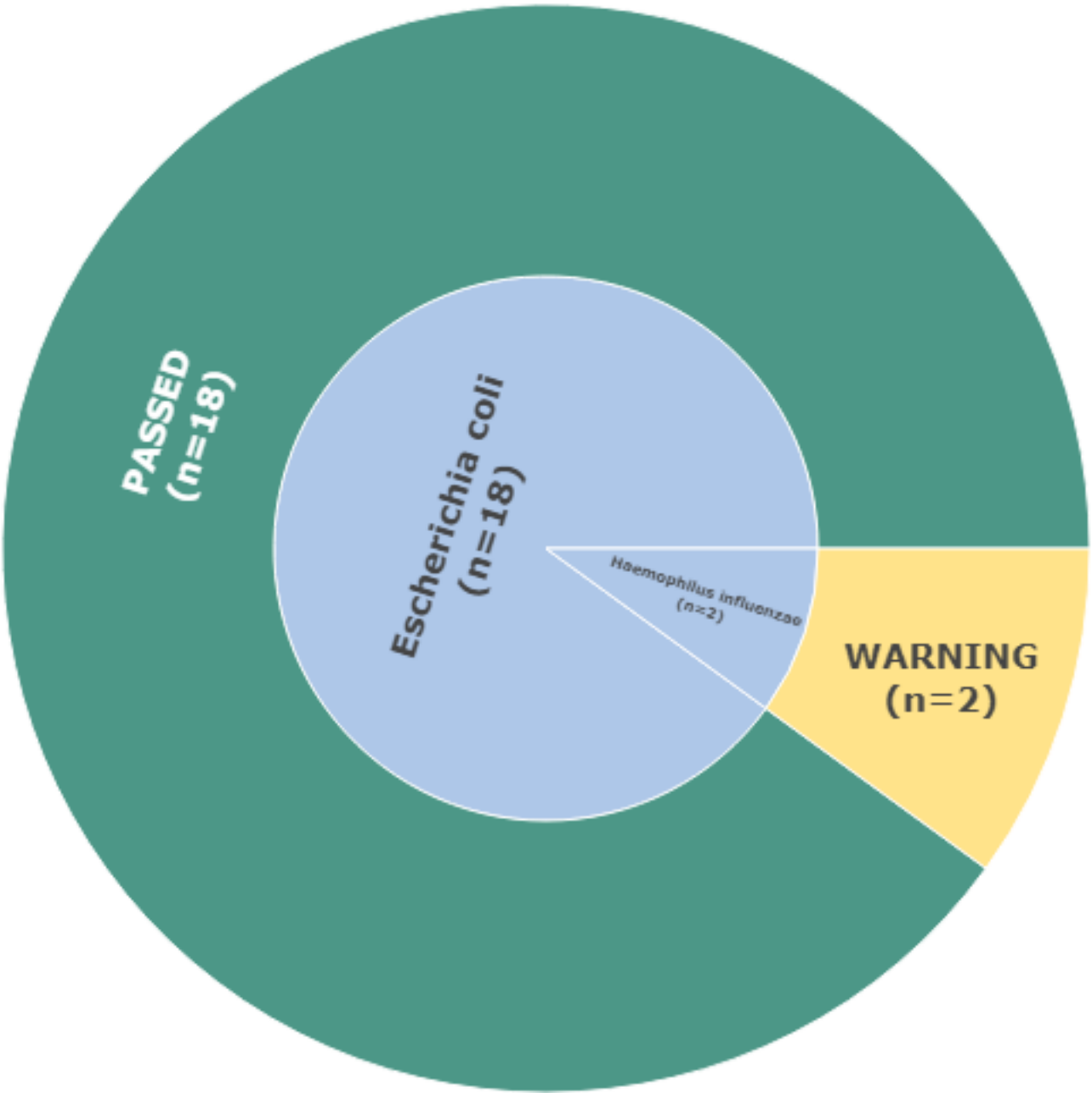


Figure 1. **Distribution and quality control results of sequenced isolates.**

Table 1. **Concordance analysis of bacterial identification of sequenced isolates.**

Isolate No.	Sample ID	Description	Index	ARSRL	WGS
1	25ARS_UTPR_BL_0003	UDI170	B10	*Escherichia coli*	*Escherichia coli*
2	25ARS_UTPR_BL_0004	UDI171	C10	*Escherichia coli*	*Escherichia coli*
3	25ARS_UTPR_BL_0005	UDI172	D10	*Escherichia coli*	*Escherichia coli*
4	26UTPR_BL_0001	UDI173	E10	*Escherichia coli*	*Escherichia coli*
5	26UTPR_BL_0002	UDI174	F10	*Escherichia coli*	*Escherichia coli*
6	26UTPR_BL_0003	UDI175	G10	*Escherichia coli*	*Escherichia coli*
7	26UTPR_BL_0004	UDI176	H10	*Escherichia coli*	*Escherichia coli*
8	26UTPR_BL_0005	UDI138	B06	*Escherichia coli*	*Escherichia coli*
9	26UTPR_BL_0006	UDI139	C06	*Escherichia coli*	*Escherichia coli*
10	25ARS_NMS0037_0001	UDI161	A09	*No Referred Data*	*Escherichia coli*
11	25ARS_NMS0041_0001	UDI162	B09	*No Referred Data*	*Escherichia coli*
12	25ARS_NMS0042_0001	UDI163	C09	*No Referred Data*	*Escherichia coli*
13	25ARS_NMS0044_0001	UDI164	D09	*No Referred Data*	*Escherichia coli*
14	25ARS_NMS0045_0001	UDI165	E09	*No Referred Data*	*Escherichia coli*
15	25ARS_NMS0047_0001	UDI166	F09	*No Referred Data*	*Escherichia coli*
16	25ARS_NMS0048_0001	UDI167	G09	*No Referred Data*	*Escherichia coli*
17	25ARS_NMS0049_0001	UDI168	H09	*No Referred Data*	*Escherichia coli*
18	25ARS_NMS0052_0001	UDI169	A10	*No Referred Data*	*Escherichia coli*
19	26ARS_BGH80OT	UDI141	E06	*No Referred Data*	*Haemophilus influenzae*
20	26ARS_JLM0121OT	UDI140	D06	*No Referred Data*	*Haemophilus influenzae*

Legend: PASS | FAILURE | (x) - NON-CONCORDANT |

RAW READS QC

Table 2. **Summary of raw reads quality control analysis.**

Isolate No.	Sample ID	Species	Species Abundance	Coverage (Qualibact)	Duplication Rate	GC Content	MLST (ST)	Read Length	Read Q30	Final Status
1	25ARS_UTPR_BL_0003	*Escherichia coli*	100	49.46	0.000455807	50.7	131	149 - 149	0.943188	PASSED
2	25ARS_UTPR_BL_0004	*Escherichia coli*	100	45.34	0.000439143	50.7	131	148 - 148	0.955170	PASSED
3	25ARS_UTPR_BL_0005	*Escherichia coli*	100	29.28	0.000825474	50.5	144	144 - 144	0.950181	PASSED
4	26UTPR_BL_0001	*Escherichia coli*	100	41.49	0.000394814	50.3	131	148 - 148	0.953506	PASSED
5	26UTPR_BL_0002	*Escherichia coli*	100	51.32	0.000544885	50.3	5229	147 - 147	0.949889	PASSED
6	26UTPR_BL_0003	*Escherichia coli*	100	45.99	0.00116872	50.4	95	143 - 143	0.946523	PASSED
7	26UTPR_BL_0004	*Escherichia coli*	100	64.18	0.000655237	50.5	1193	148 - 148	0.954130	PASSED
8	26UTPR_BL_0005	*Escherichia coli*	100	38.32	0.000572639	50.6	1193	147 - 147	0.952215	PASSED
9	26UTPR_BL_0006	*Escherichia coli*	100	42.26	0.00053697	50.7	2935	148 - 149	0.955077	PASSED
10	25ARS_NMS0037_0001	*Escherichia coli*	100	59.11	0.000620521	50.5	4388	149 - 149	0.957710	PASSED
11	25ARS_NMS0041_0001	*Escherichia coli*	100	67.88	0.000712432	50.2	354	147 - 147	0.958947	PASSED
12	25ARS_NMS0042_0001	*Escherichia coli*	100	40.84	0.000568546	49.8	155	148 - 148	0.958711	PASSED
13	25ARS_NMS0044_0001	*Escherichia coli*	100	44.56	0.000401452	50.7	1431	148 - 148	0.949488	PASSED
14	25ARS_NMS0045_0001	*Escherichia coli*	100	36.33	0.000502095	50.8	2448	145 - 145	0.942804	PASSED
15	25ARS_NMS0047_0001	*Escherichia coli*	100	32.86	0.000508602	50.0	155	146 - 146	0.951554	PASSED
16	25ARS_NMS0048_0001	*Escherichia coli*	100	51.84	0.000536762	50.4	212	148 - 148	0.949777	PASSED
17	25ARS_NMS0049_0001	*Escherichia coli*	100	30.32	0.000478915	50.9	162	146 - 146	0.942677	PASSED
18	25ARS_NMS0052_0001	*Escherichia coli*	100	56.52	0.000428025	51.4	93	150 - 150	0.937701	PASSED
19	26ARS_BGH80OT	*Haemophilus influenzae*	100	109	0.000660643	38.2		147 - 147	0.968506	WARNING
20	26ARS_JLM0121OT	*Haemophilus influenzae*	100	107.57	0.000676511	38.0		149 - 149	0.971211	WARNING

Legend: PASSED | WARNING | FAILED

Summary of Raw Reads with FAILED QC results

Sample ID	Reason - Failed Metrics
	No further action required for this batch.

ASSEMBLY QC

Table 3. **Summary of assembly reads quality control analysis.**

Isolate No.	Sample ID	WGS ID	Completeness	Contamination	Total Coding Sequences	Genome Size	GC Content	N50	Total Contig	Raw Reads Result	Assembly Result
1	25ARS_UTPR_BL_0003	*Escherichia coli*	100	0.79	4675	4.93 Mb	50.76	244527	103	PASSED	PASSED
2	25ARS_UTPR_BL_0004	*Escherichia coli*	100	0.79	4667	4.93 Mb	50.75	244527	99	PASSED	PASSED
3	25ARS_UTPR_BL_0005	*Escherichia coli*	100	0.53	4814	5.03 Mb	50.62	259749	103	PASSED	PASSED
4	26UTPR_BL_0001	*Escherichia coli*	100	1.02	5034	5.30 Mb	50.58	185622	102	PASSED	PASSED
5	26UTPR_BL_0002	*Escherichia coli*	100	0.12	4690	4.93 Mb	50.56	142756	101	PASSED	PASSED
6	26UTPR_BL_0003	*Escherichia coli*	100	0.75	5108	5.31 Mb	50.58	268695	73	PASSED	PASSED
7	26UTPR_BL_0004	*Escherichia coli*	100	1.25	5001	5.27 Mb	50.52	354351	79	PASSED	PASSED
8	26UTPR_BL_0005	*Escherichia coli*	100	0.88	4759	5.06 Mb	50.66	237307	74	PASSED	PASSED
9	26UTPR_BL_0006	*Escherichia coli*	100	0.1	4292	4.56 Mb	50.88	119539	103	PASSED	PASSED
10	25ARS_NMS0037_0001	*Escherichia coli*	100	0.71	5097	5.28 Mb	50.59	131659	115	PASSED	PASSED
11	25ARS_NMS0041_0001	*Escherichia coli*	100	1.02	4854	5.18 Mb	50.35	232514	69	PASSED	PASSED
12	25ARS_NMS0042_0001	*Escherichia coli*	100	0.54	5071	5.21 Mb	50.35	118304	142	PASSED	PASSED
13	25ARS_NMS0044_0001	*Escherichia coli*	100	0.48	4887	5.13 Mb	50.61	166804	72	PASSED	PASSED
14	25ARS_NMS0045_0001	*Escherichia coli*	100	0.51	4679	4.94 Mb	50.51	217869	60	PASSED	PASSED
15	25ARS_NMS0047_0001	*Escherichia coli*	100	0.39	4872	5.08 Mb	50.52	140138	116	PASSED	PASSED
16	25ARS_NMS0048_0001	*Escherichia coli*	100	0.27	5266	5.36 Mb	50.44	166308	143	PASSED	PASSED
17	25ARS_NMS0049_0001	*Escherichia coli*	100	0.09	4916	5.18 Mb	50.43	190924	100	PASSED	PASSED
18	25ARS_NMS0052_0001	*Escherichia coli*	100	0.47	5210	5.27 Mb	50.59	98082	267	PASSED	PASSED
19	26ARS_BGH80OT	*Haemophilus influenzae*	100	0.03	1886	1.90 Mb	38.08	124416	34	WARNING	PASSED
20	26ARS_JLM0121OT	*Haemophilus influenzae*	100	0.03	1809	1.89 Mb	37.96	275050	26	WARNING	PASSED

Legend: PASSED | WARNING | FAILED

Final Recommendation based on QC results

Sample ID	Remarks
	No further action required for this batch.

MLST RESULTS

Table 4. **Summary of Multilocus Sequence Typing (MLST) results of *a) Escherichia coli, and b) Haemophilus influenzae***

WGS ID	MLST (ST)
Escherichia coli	4388 (n= 1), 354 (n= 1), 155 (n= 2), 1431 (n= 1), 2448 (n= 1), 212 (n= 1), 162 (n= 1), 93 (n= 1), 131 (n= 3), 144 (n= 1), 5229 (n= 1), 95 (n= 1), 1193 (n= 2), 2935 (n= 1)
Haemophilus influenzae	- (n= 2)

Legend: (-) Not identified

AMR PREDICTION RESULTS

Table 5. **Summary of AMR genes present among of *a) Escherichia coli, and b) Haemophilus influenzae***

a) *Escherichia coli* (Part 1.1)

Sample ID	AMR AZITHROMYCIN/ ERYTHROMYCIN/ SPIRAMYCIN/ TELITHROMYCIN	AMR BETA-LACTAM	AMR CEPHALOSPORIN	AMR CHLORAMPHENICOL	AMR CHLORAMPHENICOL/ FLORFENICOL	AMR EFFLUX
25ARS_NMS0037_0001	mrx(A), mph(A)	blaEC	blaCTX-M-123	NA	floR	sdeY, farB, acrF, kdeA, norM, sdeY, farB, mdtM, emrD, acrF
25ARS_NMS0041_0001	NA	blaEC	NA	NA	NA	adeE, sdeY, emrD, farB, acrF, mdtM, norM, sdeY, sdeY, kdeA, farB
25ARS_NMS0042_0001	NA	blaEC, blaTEM, blaTEM	blaCTX-M-123	NA	NA	acrF, farB, mdtM, kdeA, acrF, emrD, sdeY, farB, norM, sdeY
25ARS_NMS0044_0001	NA	blaEC	blaCTX-M-15	NA	NA	mdtM, acrF, farB, sdeY, norM, sdeY, acrF, emrD, farB, kdeA
25ARS_NMS0045_0001	NA	blaEC	blaCTX-M-55	NA	NA	farB, sdeY, acrF, sdeY, acrF, mdtM, norM, farB, kdeA, emrD
25ARS_NMS0047_0001	mph(A), mrx(A)	blaEC, blaTEM-1	blaCTX-M-65	NA	NA	acrF, emrD, norM, farB, kdeA, farB, mdtM, acrF, sdeY, sdeY

a) *Escherichia coli* (Part 1.2)

Sample ID	AMR FOSFOMYCIN	AMR GENTAMICIN	AMR KAMYCIN	AMR LINCOSAMIDE	AMR QUINOLONE	AMR STREPTOMYCIN
25ARS_NMS0037_0001	NA	NA	aph(3')-Ia	NA	qnrS1	NA
25ARS_NMS0041_0001	NA	NA	NA	NA	NA	NA
25ARS_NMS0042_0001	fosA4, fosA3	NA	NA	NA	qnrS13	aadA2, aadA5
25ARS_NMS0044_0001	NA	NA	NA	NA	NA	NA
25ARS_NMS0045_0001	NA	NA	aph(3')-Ia	NA	NA	aadA1
25ARS_NMS0047_0001	fosA3	NA	NA	NA	NA	aadA5, aph(3")-Ib, aph(6)-Id

a) *Escherichia coli* (Part 1.3)

Sample ID	AMR STREPTOTHRICIN	AMR SULFOMIDE	AMR TETRACYCLINE	AMR TRIMETHOPRIM	AMR TYLOSIN	STRESS ARSENIC
25ARS_NMS0037_0001	NA	NA	tet(A)	NA	NA	arsR, arsR, arsH
25ARS_NMS0041_0001	NA	NA	tet(B)	dfrA17	estT	arsR
25ARS_NMS0042_0001	NA	sul2	tet(A)	dfrA12, dfrA17	estT	arsR
25ARS_NMS0044_0001	NA	NA	tet(A)	NA	estT	arsR
25ARS_NMS0045_0001	NA	sul3	NA	NA	estT, estT, estX-3	arsR
25ARS_NMS0047_0001	NA	sul1, sul2	tet(A)	dfrA17	estT	arsR

a) *Escherichia coli* (Part 1.4)

Sample ID	STRESS ARSENITE	STRESS ARSETE	STRESS COPPER	STRESS COPPER/ SILVER	STRESS EFFLUX	STRESS MERCURY
25ARS_NMS0037_0001	arsB, arsD, arsA, arsB, arsA	arsC, arsC	copR	silS, silR, silC, silB, silA	smdA, smdB	NA
25ARS_NMS0041_0001	arsB	arsC	copR	silA, silB, silC, silR, silS	smdA, smdB	NA
25ARS_NMS0042_0001	arsB	arsC	copR	silA, silB, silC, silR, silS	smdA, smdB, emrE	NA
25ARS_NMS0044_0001	arsB	arsC	copR	silA, silB, silC, silR, silS	smdA, smdB	NA
25ARS_NMS0045_0001	arsB	arsC	pcoE, pcoS, pcoR, pcoD, pcoC, pcoB, pcoA, copR	silS, silR, silC, silB, silA, silA, silB, silF, silC, silR, silS	smdB, smdA	NA
25ARS_NMS0047_0001	arsB	arsC	copR, pcoE, pcoS, pcoR, pcoD, pcoC, pcoB, pcoA	silB, silA, silC, silR, silS, silA, silB, silF	smdB, smdA	NA

a) *Escherichia coli* (Part 1.5)

Sample ID	STRESS NA	STRESS ORGANOMERCURY	STRESS QUATERRY AMMONIUM	STRESS SILVER	STRESS TELLURIUM	VIRULENCE NA
25ARS_NMS0037_0001	asr, fieF, ariR	NA	NA	NA	NA	steB, stcD, sty4826, lpfB, lpfA, fdeC, yigF, espC, sodC1, ymoA, gip, allD, ironN, iroE, hscC, sfaF, iroN, sfaF, sslE, staC, espX1, mltE, traT, ymoA, hlyE, katP, icmF, iss, lpfA-O113, lsrF, yfeA, yfeB, yfeD, iroD, iroC, iroB
25ARS_NMS0041_0001	fieF, ariR, asr	NA	NA	NA	NA	katP, fdeC, ymoA, gip, iha, eilA, air, lpfA-O113, ymoA, stcD, sslE, stm2239, sodC1, rhuM, staC, ibeA, gip, ibeA, sfaF, iucA, iucB, iucC, iucD, iutA, sfaF, ireA, hlyE, sodC1, lsrF, sinH, steB, hscC, ironN, allD, ymoA, yfeB, yfeD, yfeA
25ARS_NMS0042_0001	asr, ariR, fieF	NA	NA	NA	NA	hscC, ironN, espX1, stm2239, ymoA, stm2239, sfaF, ymoA, hlyE, lpfA-O113, staC, ymoA, sfaF, stcD, fdeC, mltE, traT, ymoA, steB, sslE, lsrF, katP, sodC1, gip, allD, iss
25ARS_NMS0044_0001	asr, ariR, fieF	NA	NA	NA	NA	stcD, sfaF, sty4826, hscC, ironN, allD, gip, staC, espX1, xis, int, sodC1, ymoA, lsrF, sfaF, hlyE, aafD, afaC, papC, papH, papA, fdeC, lpfA-O113, ymoA, traT, mltE, sslE, katP, vasL, tssA, ymoA, capU, steB, iss
25ARS_NMS0045_0001	asr, fieF, ariR	NA	qacL	silP, silE	NA	steB, stcD, ironN, hscC, ymoA, ymoA, allD, gip, ymoA, sfaF, sodC1, lsrF, sfaF, yfeA, yfeB, yfeD, iucA, iucB, iucC, iucD, iutA, sslE, fdeC, sty4826, espX1, staC, ironN, iroE, iroD, iroC, iroB, iss, sty4826, mltE, traT, katP, vasL, tssA, hlyE, lpfA-O113, stm2239
25ARS_NMS0047_0001	asr, ariR, fieF	NA	qacEdelta1	silE, silP	NA	lpfA-O113, sfaF, lsrF, ymoA, staC, ymoA, vasL, mltE, traT, steB, fdeC, sslE, sodC1, hlyE, stcD, sty4826, ironN, hscC, iucC, iucB, iucA, yfeD, yfeB, yfeA, katP, sfaF, ironN, iroE, iroD, iroC, allD, gip, ymoA, ymoA, iutA, iucD, espX1

a) *Escherichia coli* (Part 2.1)

	Sample ID	AMR AZITHROMYCIN/ ERYTHROMYCIN/ SPIRAMYCIN/ TELITHROMYCIN	AMR BETA-LACTAM	AMR CEPHALOSPORIN	AMR CHLORAMPHENICOL	AMR CHLORAMPHENICOL/ FLORFENICOL	AMR EFFLUX
7	25ARS_NMS0048_0001	NA	blaEC	blaDHA-1	NA	floR	sdeY, farB, kdeA, mdtM, farB, norM, acrF, emrD, acrF, sdeY
8	25ARS_NMS0049_0001	mrx(A), mph(A)	blaTEM-1, blaEC	blaCTX-M-123	NA	NA	acrF, sdeY, emrD, sdeY, farB, farB, norM, acrF, mdtM, kdeA
9	25ARS_NMS0052_0001	mph(A), mrx(A)	blaEC	blaCTX-M-123	NA	NA	mdtM, acrF, sdeY, kdeA, emrD, farB, farB, sdeY, norM, acrF
10	25ARS_UTPR_BL_0003	mrx(A), mph(A)	blaEC	blaCTX-M-15, blaDHA-1	catA1	NA	kdeA, sdeY, emrD, farB, sdeY, mdtM, norM, acrF, adeE, farB, sdeY
11	25ARS_UTPR_BL_0004	mrx(A), mph(A)	blaEC	blaCTX-M-15, blaDHA-1	catA1	NA	kdeA, adeE, sdeY, farB, emrD, sdeY, farB, sdeY, acrF, mdtM, norM
12	25ARS_UTPR_BL_0005	NA	blaEC, blaTEM-1	NA	NA	NA	sdeY, sdeY, acrF, norM, farB, kdeA, emrD, adeE, acrF, mdtM, farB

a) *Escherichia coli* (Part 2.2)

	Sample ID	AMR FOSFOMYCIN	AMR GENTAMICIN	AMR KAMYCIN	AMR LINCOSAMIDE	AMR QUINOLONE	AMR STREPTOMYCIN
7	25ARS_NMS0048_0001	NA	NA	NA	NA	qnrB4	NA
8	25ARS_NMS0049_0001	fosA3	NA	NA	NA	NA	aph(6)-Id
9	25ARS_NMS0052_0001	fosA3	NA	aph(3')-Ia	NA	NA	aadA1, aph(6)-Id, aph(3")-Ib
10	25ARS_UTPR_BL_0003	NA	NA	NA	NA	qnrB4	NA
11	25ARS_UTPR_BL_0004	NA	NA	NA	NA	qnrB4	NA
12	25ARS_UTPR_BL_0005	NA	NA	NA	NA	NA	NA

a) *Escherichia coli* (Part 2.3)

	Sample ID	AMR STREPTOTHRICIN	AMR SULFOMIDE	AMR TETRACYCLINE	AMR TRIMETHOPRIM	AMR TYLOSIN	STRESS ARSENIC
7	25ARS_NMS0048_0001	NA	sul1	tet(B)	NA	estT	arsR, arsR
8	25ARS_NMS0049_0001	NA	NA	tet(A)	NA	estX	arsR
9	25ARS_NMS0052_0001	sat2	sul2	NA	NA	estX	arsR
10	25ARS_UTPR_BL_0003	NA	sul1	NA	dfrA17	NA	NA
11	25ARS_UTPR_BL_0004	NA	sul1	NA	dfrA17	NA	NA
12	25ARS_UTPR_BL_0005	NA	NA	NA	NA	NA	NA

a) *Escherichia coli* (Part 2.4)

	Sample ID	STRESS ARSENITE	STRESS ARSETE	STRESS COPPER	STRESS COPPER/ SILVER	STRESS EFFLUX	STRESS MERCURY
7	25ARS_NMS0048_0001	arsD, arsB	arsC	copR	silC, silR, silS, silA, silB	smdB, smdA, emrE, emrE	merP, merT, merR, merE, merD, merA
8	25ARS_NMS0049_0001	arsB	arsC	copR	silC, silA, silB, silR, silS	smdB, smdA, emrE, emrE	NA
9	25ARS_NMS0052_0001	arsB	arsC	copR	silA, silS, silR, silC, silB	emrE, smdA, smdB, mtrF	NA
10	25ARS_UTPR_BL_0003	NA	arsC	copR	silC, silB, silA, silS, silR	smdB, smdA, emrE, emrE	NA
11	25ARS_UTPR_BL_0004	NA	arsC	copR	silA, silB, silC, silR, silS	smdA, smdB, emrE, emrE	NA
12	25ARS_UTPR_BL_0005	NA	arsC	copR	silC, silB, silS, silR, silA	smdB, smdA, emrE	NA

a) *Escherichia coli* (Part 2.5)

	Sample ID	STRESS NA	STRESS ORGANOMERCURY	STRESS QUATERRY AMMONIUM	STRESS SILVER	STRESS TELLURIUM	VIRULENCE NA
7	25ARS_NMS0048_0001	asr, ariR, fieF, terA	merC	qacE	NA	terE, terD, terC, terB, terZ, terW	steB, stcD, traT, espX1, fdeC, staC, sfaF, hscC, iroN, tssA, ymoA, capU, mltE, iss, sfaF, lsrF, sodC1, hlyE, lpfB, lpfA, katP, sslE, mltE, vasL, sodC1, lpfA-O113, allD, gip, ymoA
8	25ARS_NMS0049_0001	asr, ariR, fieF	NA	NA	NA	NA	mchF, iucD, iucC, iucB, iucA, yfeD, yfeB, yfeA, astA, aafD, afaC, iutA, sodC1, int, xis, steB, stcD, lpfA-O113, sfaF, lsrF, ymoA, iroN, hscC, fdeC, allD, gip, hlyE, lpfB, lpfA, mltE, traT, sslE, papC, papH, ymoA, katP, tssA, vasL, iss, sfaF, staC, espX1
9	25ARS_NMS0052_0001	fieF, asr	NA	NA	NA	NA	sslE, astA, gip, stcD, yigF, iha, etpD, sodC1, iha, allD, capU, hscC, ymoA, hlyE, yfeA, yfeB, yfeD, iutA, iucD, iucC, iucB, steB, astA, katP, ymoA, vasL, iroN, lsrF, sfaF, tsh, sodC1, iss, stm2239, iucA, espX1, papG-II, papF, papE, papA, papH, papC
10	25ARS_UTPR_BL_0003	fieF, ariR, asr	NA	qacE	NA	NA	iroN, hscC, rhuM, katP, yigF, stcD, fdeC, ymoA, sfaF, staC, iha, ybtQ, ybtP, steB, papA, espC, etpD, allD, gip, ymoA, ymoA, sslE, iucA, iucB, iucC, iucD, iutA, sat, traT, mltE, yfeA, yfeB, yfeD, sinH, sodC1, sfaF
11	25ARS_UTPR_BL_0004	asr, ariR, fieF	NA	qacE	NA	NA	hscC, sfaF, yfeD, yfeB, yfeA, sinH, sat, iutA, iucD, iucC, iucB, iucA, traT, sfaF, mltE, espC, iroN, rhuM, ymoA, katP, yigF, sslE, ymoA, fdeC, etpD, gip, allD, stcD, iha, ymoA, steB, staC, papA, ybtP, ybtQ, sodC1
12	25ARS_UTPR_BL_0005	asr, ariR, fieF	NA	NA	NA	NA	gip, allD, ymoA, fdeC, sinH, stcD, ybtP, ybtQ, sodC1, bfpA, etpD, vactox, tssA, vasL, iucA, iucB, iucC, iucD, iutA, iroN, rhuM, katP, iha, papA, lpfA, lpfB, yfeD, yfeB, yfeA, sfaF, mltE, traT, ymoA, stm2239, int, papE, papC, papH, sat, sslE, senB, papA, staC, sfaF, papG-II, papF, int, xis, stm2239, steB

a) *Escherichia coli* (Part 3.1)

	Sample ID	AMR AZITHROMYCIN/ ERYTHROMYCIN/ SPIRAMYCIN/ TELITHROMYCIN	AMR BETA-LACTAM	AMR CEPHALOSPORIN	AMR CHLORAMPHENICOL	AMR CHLORAMPHENICOL/ FLORFENICOL	AMR EFFLUX
13	26UTPR_BL_0001	mph(A), mrx(A)	blaEC, blaTEM-1	NA	catA1	NA	farB, kdeA, emrD, adeE, farB, acrF, sdeY, norM, acrF, sdeY
14	26UTPR_BL_0002	NA	blaEC, blaTEM-1	NA	cmlA1	NA	sdeY, farB, sdeY, emrD, mdtM, kdeA, sdeY, farB, acrF, norM
15	26UTPR_BL_0003	NA	blaTEM-1, blaEC	NA	NA	NA	kdeA, acrF, sdeY, adeE, norM, farB, sdeY, farB, sdeY, emrD
16	26UTPR_BL_0004	NA	blaTEM-1	blaEC-5, blaCTX-M-15	NA	NA	adeE, sdeY, kdeA, farB, farB, sdeY, acrF, norM, emrD, sdeY
17	26UTPR_BL_0005	NA	NA	blaEC-5	NA	NA	kdeA, sdeY, adeE, acrF, sdeY, farB, sdeY, farB, norM, emrD
18	26UTPR_BL_0006	NA	blaEC	NA	NA	NA	sdeY, farB, emrD, farB, kdeA, norM, mdtM, sdeY

a) *Escherichia coli* (Part 3.2)

	Sample ID	AMR FOSFOMYCIN	AMR GENTAMICIN	AMR KAMYCIN	AMR LINCOSAMIDE	AMR QUINOLONE	AMR STREPTOMYCIN
13	26UTPR_BL_0001	NA	aac(3)-IId	NA	NA	NA	NA
14	26UTPR_BL_0002	NA	NA	aph(3')-Ia	lnu(F)	NA	aadA2, aadA1, aadA2
15	26UTPR_BL_0003	fosA4	NA	aph(3')-Ia	NA	qnrS13	aadA1, aph(6)-Id
16	26UTPR_BL_0004	NA	NA	NA	NA	NA	NA
17	26UTPR_BL_0005	NA	NA	NA	NA	NA	NA
18	26UTPR_BL_0006	NA	NA	NA	NA	NA	NA

a) *Escherichia coli* (Part 3.3)

	Sample ID	AMR STREPTOTHRICIN	AMR SULFOMIDE	AMR TETRACYCLINE	AMR TRIMETHOPRIM	AMR TYLOSIN	STRESS ARSENIC
13	26UTPR_BL_0001	NA	sul1	NA	NA	NA	NA
14	26UTPR_BL_0002	NA	sul2, sul3	tet(M)	dfrA12	NA	arsR
15	26UTPR_BL_0003	NA	sul3	tet(A)	dfrA14	estX-3	NA
16	26UTPR_BL_0004	NA	NA	NA	dfrA17	NA	NA
17	26UTPR_BL_0005	NA	NA	tet(B)	dfrA17	NA	NA
18	26UTPR_BL_0006	NA	NA	NA	NA	NA	arsR

a) *Escherichia coli* (Part 3.4)

	Sample ID	STRESS ARSENITE	STRESS ARSETE	STRESS COPPER	STRESS COPPER/ SILVER	STRESS EFFLUX	STRESS MERCURY
13	26UTPR_BL_0001	NA	arsC	copR	silR, silS, silA, silB, silC	emrE, smdA, smdB	NA
14	26UTPR_BL_0002	arsB	arsC	pcoE, pcoS, pcoR, pcoD, pcoC, pcoB, pcoA, copR	silB, silS, silF, silC, silR, silA, silS, silR, silC, silB, silA	emrE, smdB, smdA	NA
15	26UTPR_BL_0003	NA	arsC	copR	silC, silB, silA, silS, silR	emrE, smdA, smdB	NA
16	26UTPR_BL_0004	NA	arsC	copR	silB, silA, silS, silR, silC	smdA, smdB, emrE	NA
17	26UTPR_BL_0005	NA	arsC	copR	silA, silB, silC, silR, silS	smdB, smdA, emrE	merR, merT, merP, merA, merD, merE
18	26UTPR_BL_0006	arsB	arsC	copR	silA, silB, silC, silR, silS	smdB, smdA	NA

a) *Escherichia coli* (Part 3.5)

	Sample ID	STRESS NA	STRESS ORGANOMERCURY	STRESS QUATERRY AMMONIUM	STRESS SILVER	STRESS TELLURIUM	VIRULENCE NA
13	26UTPR_BL_0001	fieF, ariR, asr	NA	NA	NA	NA	sslE, katP, sfaF, yfeA, yfeB, yfeD, iss, ymoA, ybtP, yigF, steB, iroN, hscC, aafB, afaC, aafD, nfaE, ymoA, aggR, aap, aar, aatA, iutA, iucD, iucC, iucB, iucA, fdeC, ymoA, agg3D, nfaE, ybtQ, staC, allD, gip, traT, mltE, sfaF, sodC1, etpD, sty3281, sinH, espC, hlyA-alpha, cnf1, f17a, papA, papH, papC, papE, agg3C, agg3B, agg5A, papG-II, papF
14	26UTPR_BL_0002	fieF, ariR, asr	NA	qacL	silE, silP	NA	lpfB, lpfA, sfaF, stcD, ymoA, lpfA-O113, sfaF, ymoA, gip, allD, iroN, ymoA, fdeC, hscC, papH, papC, stm2239, lsrF, katP, hlyE, sodC1, staC, espX1, steB
15	26UTPR_BL_0003	fieF, ariR, asr	NA	qacL	NA	NA	steB, iss, stm2239, tsh, yfeD, iucA, iucB, iucC, iucD, iutA, staC, yfeB, vasL, sty4826, stcD, sslE, ireA, papA, papH, papC, papE, papF, papG-II, iss, iroB, iroC, iroD, iroE, iroN, cvaB, cvaC, katP, ybtP, ybtQ, stm2239, yfeA, sfaF, sodC1, traT, mltE, etpD, stm2239, tssA, iroN, vactox, fdeC, ymoA, sfaF, rhuM, gip, allD
16	26UTPR_BL_0004	ariR, asr, fieF	NA	NA	NA	NA	f17a, sfaF, sinH, ymoA, yfeB, yfeD, iroN, fdeC, vasL, tssA, gip, allD, btpA, ybtQ, ybtP, stcD, steB, yfeA, staC, sfaF, vactox, sodC1, sslE, sty4826, katP, iucA, iucB, iucC, iucD, iutA, sat, rhuM, lpfA1, lpfB, papA, senB, iha, etpD
17	26UTPR_BL_0005	ariR, fieF, asr	merC	NA	NA	NA	iroN, ymoA, allD, gip, etpD, katP, sat, iutA, iucD, iucC, iucB, iucA, papA, staC, sfaF, senB, iha, lpfB, yfeA, yfeB, yfeD, vactox, tssA, vasL, sfaF, stcD, steB, sinH, sty4826, fdeC, sodC1, btpA, ybtQ, ybtP, clbN, clbB, lpfA1, rhuM, sslE
18	26UTPR_BL_0006	fieF, asr, ariR	NA	NA	NA	NA	espX1, iroN, hscC, katP, lsrF, stm2239, stcD, steB, yigF, sslE, etpD, sty4826, sodC1, sfaF, hlyE, ymoA, fdeC, gip, allD

b) *Haemophilus influenzae* (Part 1.1)

Sample ID	AMR AZITHROMYCIN/ ERYTHROMYCIN	AMR AZITHROMYCIN/ ERYTHROMYCIN/ STREPTOGRAMIN B	AMR BETA-LACTAM	AMR EFFLUX	VIRULENCE NA
26ARS_BGH80OT	NA	NA	NA	farB, hmrM	yfeB, yfeA, yfeD
26ARS_JLM0121OT	mef(A)	msr(D)	blaTEM-1	farB, hmrM	yfeA, lsrF, yfeB, yfeD